Ch 5.2: The Bootstrap

Lecture 14 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

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Announcements

Last time:

- k-fold CV for Classification

Announcements:

- Fifth homework posted
- Check schedule

Covered in this lecture

- Bootstrap

Section 1

The Bootstrap

The Idea

The goal: quantify the uncertainty associated with a given estimator or statistical learning method.

Ex) Could be used to estimate the standard error of linear reg. coefficients

Won't because other methods

Power of the bootstrap:

Very general method

easily applied to many stat. learning methods

Useful when no access to theory (true variance) but can simulate emperically.

The boostrap idiom

"Pull yourself up by your bootstraps."

- Originally used as a saying meaning an impossible task, used sarcastically
- Now often used to imply that socioeconomic advancement is something everyone should be able to do

Today's class: Bootstrap on a simple modeling problem

- We wish to invest a fixed sum of money in two financial assets that yield returns of $\boxed{X}$ and $\boxed{Y}$, respectively, where X and Y are random quantities.

Warning: not learning Y

- We will invest a fraction α of our money in X, and will invest the remaining $1 - \alpha$ in Y.

$\alpha \in [0,1]$

Money I make:

$\alpha X + (1-\alpha) Y$

- Since there is variability associated with the returns on these two assets, we wish to choose α to minimize the total risk, or variance, of our investment.

Minimize risk:

$Var(\alpha X + (1-\alpha) Y)$

One can show..... ⇒ You will do on homework

...that $\mathrm{Var}(\alpha X + (1-\alpha)Y)$
is minimzed by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

where

- $\sigma_X^2 = \mathrm{Var}(X)$
- $\sigma_Y^2 = \mathrm{Var}(Y)$
- $\sigma_{XY} = \mathrm{Cov}(X, Y)$

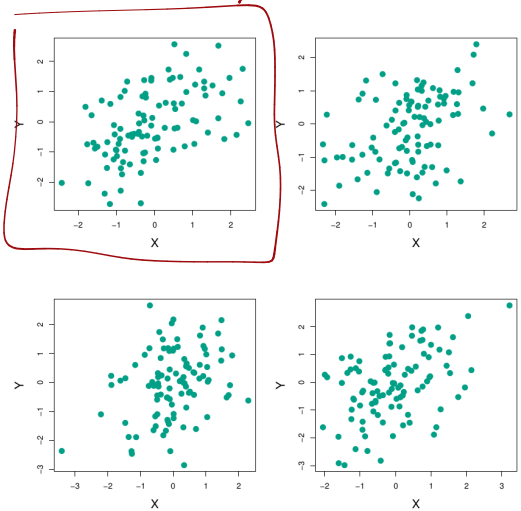
Get an estimate:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

Analyze using bootstrapping

Simulated data

$\hat{\alpha} = 0.576$



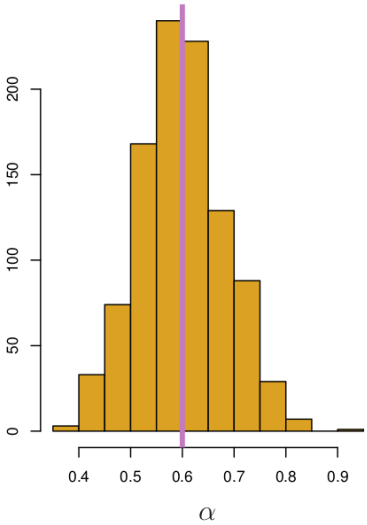
- Simulate data using
  - $\sigma_X^2 = 1$
  - $\sigma_Y^2 = 1.25$
  - $\sigma_{XY} = 0.5$
  - Implies $\alpha = 0.6$
- In each panel: Simulate 100 pairs of returns for investments

  # data pts = n
- Predict σ_X^2, σ_Y^2, and σ_{XY}
- $\hat{\alpha}$ prediction by panel:

  0.576 0.543

  0.657 0.651

Resimulate: Rinse and repeat



- Simulate 100 data points 1,000 times
- Left: Histogram of predictions for α
- Pink line: True value for α
- Mean over simulated values: $0.5996 = \bar{\alpha} = \frac{1}{1000} \sum \hat{\alpha}_r$
- St dev: $0.083 = \sqrt{\frac{1}{1000-1} \sum (\hat{\alpha}_r - \bar{\alpha})}$

Coding part 1: Apprximate $\hat{\alpha}$ with simulated data

So what's the problem?

I usually can't simulate new data sets.

Plan:

Use given data set

Create simulations from that data set

The solution: Sample the data with replacement

This is Bootstrapping

Z^{*i}

Obs	X	Y
1	4.3	2.4
2	2.1	1.1
3	5.3	2.8

Original Data (Z)

n=3

Z^{*1}

Obs	X	Y
3	5.3	2.8
1	4.3	2.4
3	5.3	2.8

n=3

$\hat{\alpha}^{*1}$

Z^{*2}

Obs	X	Y
2	2.1	1.1
3	5.3	2.8
1	4.3	2.4

n=3

$\hat{\alpha}^{*2}$

Z^{*B}

Obs	X	Y
2	2.1	1.1
2	2.1	1.1
1	4.3	2.4

n=3

$\hat{\alpha}^{*B}$

- Sample n data pts
- W/ Replacement:

  a point could show up never or multiple times

Computation of error

- Repeate procedure B times:
- Get B bootstrap data sets, $Z^{*1}, Z^{*2}, \ldots, Z^{*B}$
- Get B bootstrap estimates $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \ldots, \hat{\alpha}^{*B}$

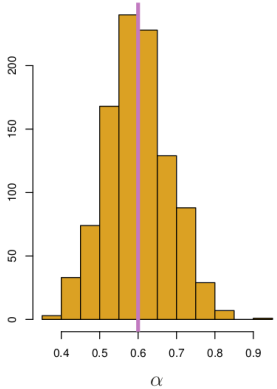
Get standard error estimate:

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \underbrace{\frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}}_{\text{average over } \hat{\alpha}^{*1}, \ldots, \hat{\alpha}^{*B}}\right)^2}$$

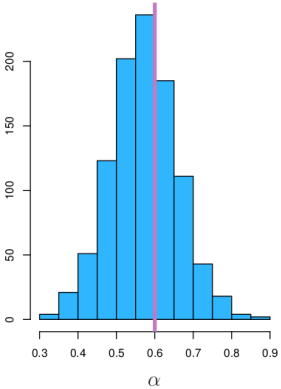
$$\sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \overline{\hat{\alpha}^{*}}\right)^2}$$

Coding part 2: Resampling data

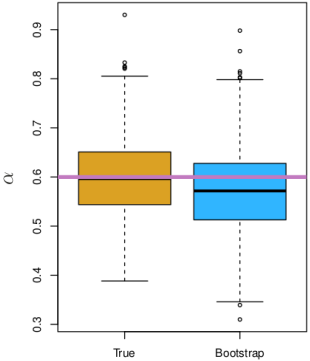
Back to the example



Resample version
Predicted $SE(\hat{\alpha}) = 0.083$

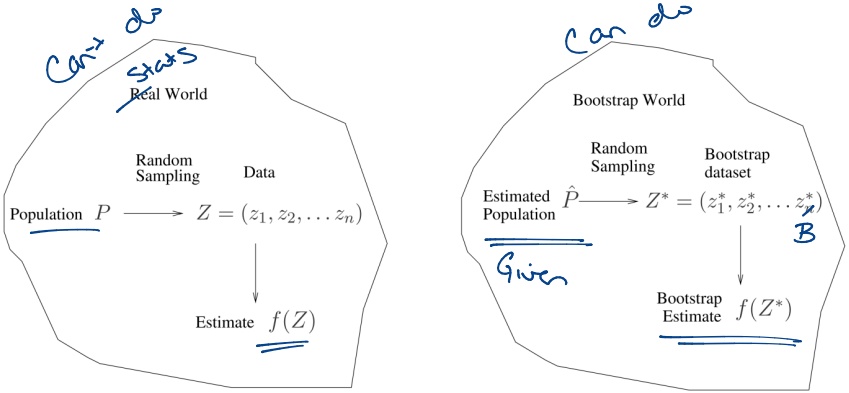


Bootstrap version
Predicted $SE(\hat{\alpha}) = 0.087$



Spreads approx the same :
Good est mate for std dev

A general picture for the bootstrap



TL;DR

- Start with data set of n points
- Sample n points with replacement to get data set Z^{*1}
- Use this to estimate whatever parameter we want $\hat{T}^{*1}$
- Repeat B times to get estimates $\hat{T}^{*1}, \ldots, \hat{T}^{*B}$
- Estimate standard error of our T estimate by

$$SE_B(\hat{T}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{T}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{T}^{*r'}\right)^2}$$ = Std dev

Boostrap vs Cross-Validation

Bootstrap:

- Resamples with replacement
- Same number of data pts per sample ~> n
- Randomness: doing this B times
- Goal: establish emperical dist. for a given statistic

k-fold **CV:**

- Take subsets without replacement
- Subsets have <n pts
- randomness: first shuffle, then subsets hit all points
- Goal: Measuring performance of a given model

Next time

10	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
11	W	Oct 5	k-fold CV	5.1.3	
12	F	Oct 7	More k-fold CV,	5.1.4-5	
13	M	Oct 10	k-fold CV for classification	5.1.5	HW #4 Due
14	W	Oct 12	Resampling methods: Bootstrap	5.2	
15	F	Oct 14	Subset selection	6.1	
16	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
17	W	Oct 19	Shrinkage: Lasso	6.2.2	
18	F	Oct 21	[No class, Dr Munch out of town]		
	M	Oct 24	No class - Fall break		
19	W	Oct 26	Dimension Reduction	6.3	
20	F	Oct 28	More dimension reduction; High dimensions	6.4	HW #6 Due
	M	Oct 31	Review		
	W	Nov 2	***Midterm #2***		